

Design Strategies for Zero-Trust Architecture in Enterprise Software: A Systematic Literature Review

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Abstract—Zero-trust architecture (ZTA) has emerged as a critical security paradigm in enterprise software development, fundamentally challenging the traditional perimeter-based security model. The main aim of this paper is to explore different design strategies for implementing zero-trust architecture in enterprise environments. However, we also explore the challenges of implementing zero-trust principles and future directions of zero-trust architecture research. We investigate the chronological distribution of relevant studies, highly cited studies, and influential authors in the field of ZTA. We conducted a systematic literature review (SLR) by thoroughly examining the area of zero-trust architecture design strategies. We find 42 pertinent studies published between 2018 and 2024 that answer our 4 research questions. The results of this SLR point out the additional research areas of zero-trust architecture implementation. However, the findings reveal the challenges faced by enterprise architects and security professionals related to implementing zero-trust principles in complex enterprise environments. Although it presents all the demographic characteristics of selected studies of ZTA. The findings of this study highlight the most common design strategies for zero-trust architecture, as well as their use in academia and industry. Furthermore, we have learnt that continuous authentication mechanisms, micro-segmentation complexity, and legacy system integration are the limitations and future gaps.

Index Terms—Zero-Trust Architecture, Network Security, Enterprise Software, Cybersecurity, Identity and Access Management, Micro-segmentation

I. INTRODUCTION

Zero-trust architecture represents a paradigm shift in cybersecurity strategy, operating on the principle of “never trust, always verify.” Unlike traditional security models that assume everything inside an organization’s network is trustworthy, zero-trust treats all network traffic as potentially hostile, regardless of its origin. This approach has become increasingly critical as organizations face sophisticated cyber threats, adopt cloud services, and support remote workforces.

The traditional perimeter-based security model, often described as a “castle and moat” approach, has proven inadequate in modern enterprise environments [1]. With the proliferation of cloud computing, mobile devices, and remote access requirements, the network perimeter has effectively dissolved.

Zero-trust architecture addresses these challenges by implementing strict identity verification, least-privilege access, and continuous monitoring across all resources.

Enterprise software systems face unique challenges in implementing zero-trust principles. These systems often involve complex integrations, legacy infrastructure, and diverse user populations with varying access requirements. Security architects must balance robust security controls with operational efficiency and user experience [2]. The implementation of zero-trust architecture requires fundamental changes to network design, identity management, access control mechanisms, and monitoring systems.

Zero-trust architecture is built upon several core principles: verify explicitly using all available data points including user identity, location, device health, and resource classification; use least privilege access by limiting user access with Just-In-Time and Just-Enough-Access policies; and assume breach by minimizing blast radius and segmenting access while verifying end-to-end encryption [3]. These principles necessitate comprehensive design strategies that address authentication, authorization, network segmentation, data protection, and threat detection.

The complexity of implementing zero-trust in enterprise environments has led to numerous research initiatives and practical frameworks. Organizations like NIST, Gartner, and Forrester have developed guidelines for zero-trust adoption, while technology vendors have created solutions addressing specific aspects of the architecture [4]. However, enterprises continue to face significant challenges including integration complexity, performance concerns, organizational resistance, and unclear return on investment.

We present a systematic literature review in the field of zero-trust architecture design strategies in the context of enterprise software. We used a structured approach to literature evaluation and selection in our review, letting the fundamental tenets of the systematic literature review procedure serve as our guide. Our review’s goals are to give a summary of recent work in software security and enterprise architecture in this field and to effectively understand the various strategies for

implementing zero-trust principles at different architectural layers. We categorize all ZTA design strategies, implementation approaches, challenges, and future directions in a model that summarizes our findings.

II. METHODOLOGY

A systematic literature review (SLR) aims to locate, assess, and interpret the research that is already available in a given area of interest. The thoroughness of the assessment search must be guaranteed by this search plan. Due to the evolving character of this systematic review, a baseline study that is comparable is required. There are many steps in this review process that must be completed in a methodical and orderly manner. These include creating a protocol for a systematic review, conducting the review in accordance with the protocol, analyzing and visualizing the results, and discussing the conclusions. We will perform this research according to Kitchenham's SLR guidelines [5].

A. Research Questions

The primary research questions of this SLR are as follows:

TABLE I
RESEARCH QUESTIONS

RQ #	Research Questions
RQ1	What design strategies and architectural patterns are available for implementing zero-trust architecture in enterprise software?
RQ2	What kind of challenges are faced by enterprise architects and security professionals related to implementing zero-trust architecture?
RQ3	What are the future directions and research gaps in zero-trust architecture?
RQ4	What are the key components and technologies used in zero-trust architecture implementations?

B. Data Sources

Six electronic databases were used as basic data sources for related studies. Moreover, we have extracted some studies from manual channels also. These electronic databases are listed in Table II.

TABLE II
ELECTRONIC DATABASES

ID	Database Name	URL
ED1	ACM	https://dl.acm.org/
ED2	IEEE Xplore	https://ieeexplore.ieee.org/
ED3	Science Direct	https://www.sciencedirect.com/
ED4	Springer Link	https://link.springer.com
ED5	Wiley	https://onlinelibrary.wiley.com/
ED6	Google Scholar	https://scholar.google.com/

C. Study Selection Procedure

This study represents an emerging field in enterprise security architecture. We needed to include studies from manual channels including Google Scholar to ensure comprehensive coverage of recent developments and industry implementations.

D. Data Extraction Form

We extracted data from 42 selected studies by using a structured data extraction form. The form included fields for: study ID, title, authors, publication year, venue, research methodology, design strategies discussed, challenges identified, technologies mentioned, validation methods, and key findings.

E. Search String

We have used population, intervention, comparison and outcome (PICO) viewpoints to construct a generic search string to keep search consistency across various electronic databases.

Search String:

("Zero Trust" OR "Zero-Trust" OR "ZTA" OR "Zero Trust Architecture") AND ("Design" OR "Architecture" OR "Strategy" OR "Framework" OR "Implementation" OR "Pattern") AND ("Enterprise" OR "Organization" OR "Corporate" OR "Business") AND ("Software" OR "System" OR "Application" OR "Network" OR "Security")

The PICO related terms for this SLR are:

- **Population:** Enterprise architects, Security professionals, Software engineers
- **Intervention:** Zero-trust design strategies, architectural patterns, security frameworks
- **Outcome:** Secure enterprise software systems, reduced attack surface, improved security posture

F. Inclusion And Exclusion Criteria

Table IV lists the inclusion and exclusion criteria used in this SLR to identify studies from the data sources that may be pertinent to the RQ. All studies obtained during the selection method fulfill these criteria.

G. Quality Assessment Criteria

The quality assessment criteria employed in this SLR is given in Table V.

III. RESULT AND DISCUSSION

All of the research questions are addressed in this section by analyzing the findings drawn from our data.

A. RQ1: Design Strategies And Architectural Patterns

What design strategies and architectural patterns are available for implementing zero-trust architecture in enterprise software?

We analyzed 42 selected primary studies and identified twelve (12) significant design strategies for zero-trust architecture in enterprise software, which are shown in Table VI. The majority of 29% papers from the 42 selected studies focus on micro-segmentation strategies, 21% papers discuss identity-centric architecture approaches, and 17% papers focus on software-defined perimeter implementations. 12% papers

TABLE III
STUDY SELECTION PROCEDURE

ID	Database Name	Initial Result Of Automated Search	Screen Result (ICs And ECs)	Final Selected Studies
ED1	ACM	15,420	289	06
ED2	IEEE Xplore	42,380	421	18
ED3	Science Direct	89,650	1,247	05
ED4	Springer Link	34,720	198	07
ED5	Wiley	28,940	156	02
ED6	Google Scholar	67,800	312	04
Total		278,910	2,623	42

TABLE IV
INCLUSION AND EXCLUSION CRITERIA

Inclusion Criteria
IC1: Articles that are peer reviewed
IC2: Articles providing information of zero-trust design strategies in context of enterprise software
IC3: Include latest study in case of multiple studies on the same theme
IC4: Articles published in last seven years (2018-2024)
IC5: Articles published in Conference, Journal and book chapter proceeding conference
IC6: Articles that provide implementation details or architectural patterns

Exclusion Criteria
EC1: Studies other than English language
EC2: Full text inaccessible articles
EC3: Short paper survey, editorials, thesis, survey paper, literature review, book chapters
EC4: Articles with weak evaluation
EC5: Articles with no validation of proposed strategies and approaches
EC6: Articles providing unclear results or findings
EC7: Articles focused solely on theoretical concepts without practical application

discuss policy-based access control, 10% papers address continuous verification mechanisms, and 8% papers cover least privilege access strategies. The remaining 3% includes API gateway patterns, service mesh architecture, data-centric security, device trust architecture, ZTNA, and behavioral analytics integration.

From our analysis, we observed that micro-segmentation is the most widely discussed design strategy for zero-trust architecture. This approach enables organizations to create granular security zones, limiting lateral movement of threats and reducing the blast radius of potential breaches. Identity-centric architecture emerged as the second most prevalent strategy, emphasizing that identity has become the new security perimeter in modern enterprise environments. Software-defined perimeter follows as the third most common approach, enabling dynamic and context-aware access control that adapts to changing threat landscapes.

B. RQ2: Implementation Challenges

What kind of challenges are faced by enterprise architects and security professionals related to implementing zero-trust architecture?

TABLE V
QUALITY ASSESSMENT CRITERIA

Quality Assessment Criteria
QC1: Context and objectives are clearly specified
QC2: The research design and conclusion support the objectives of the study
QC3: A clear statement of limitation and contribution
QC4: There should be minimal biasness in the study
QC5: Implementation or validation methodology is clearly described
QC6: Results are reproducible and verifiable

Zero-trust architecture implementation involves significant organizational, technical, and operational challenges. While analyzing ZTA implementations across enterprise environments, architects and security professionals face numerous obstacles:

- 1) **Legacy System Integration Challenge:** Enterprise architects face significant difficulties integrating zero-trust principles with legacy applications and infrastructure that were designed with implicit trust assumptions. Many legacy systems lack modern authentication mechanisms, API interfaces, or the ability to support fine-grained access controls.
- 2) **Performance and Latency Concerns:** The continuous verification and encryption requirements of zero-trust architecture can introduce latency in application performance. Security professionals must balance security requirements with user experience and application performance expectations.
- 3) **Complexity in Policy Management:** Managing access policies across diverse user populations, applications, and resources creates significant complexity. Organizations struggle with policy conflicts, maintaining consistency, and ensuring policies remain current as the environment evolves.
- 4) **Cultural and Organizational Resistance:** Zero-trust implementation requires significant changes to operational processes and user behaviors. Security professionals face resistance from users accustomed to more permissive access models and from IT teams concerned about increased operational overhead.
- 5) **Visibility and Monitoring Gaps:** Implementing comprehensive visibility across all network traffic, user activities, and resource access patterns requires significant

TABLE VI
ZERO-TRUST ARCHITECTURE DESIGN STRATEGIES AND PATTERNS

Sr. No	Strategy/Pattern	Description	Supported Security Activity	Ref
1	Identity-Centric Architecture	Implements strong identity verification as the primary security perimeter using multi-factor authentication, biometric verification, and continuous authentication mechanisms	Authentication, Authorization	[6]–[8]
2	Micro-segmentation	Divides the network into small zones to maintain separate access for separate parts of the network, limiting lateral movement of threats	Network Security, Access Control	[9]–[12]
3	Software-Defined Perimeter (SDP)	Creates dynamically provisioned perimeters around individual resources or groups of resources based on identity and context	Network Architecture, Access Control	[?], [?], [?]
4	Policy-Based Access Control	Implements dynamic access policies based on user identity, device posture, location, and risk assessment	Authorization, Risk Management	[?], [?], [?]
5	Continuous Verification	Establishes ongoing authentication and authorization checks throughout user sessions rather than one-time verification	Authentication, Monitoring	[?], [?]
6	Least Privilege Access	Implements Just-In-Time (JIT) and Just-Enough-Access (JEA) principles to minimize access rights and exposure window	Authorization, Access Control	[?], [?], [?]
7	API Gateway Pattern	Centralizes API security enforcement with authentication, authorization, rate limiting, and threat detection	API Security, Integration	[?], [?]
8	Service Mesh Architecture	Implements zero-trust principles at the service-to-service communication layer with mutual TLS and policy enforcement	Service Security, Encryption	[?], [?], [?]
9	Data-Centric Security	Focuses on protecting data itself through encryption, tokenization, and rights management regardless of location	Data Protection, Encryption	[?], [?]
10	Device Trust Architecture	Validates device health and compliance status before granting access to enterprise resources	Device Management, Access Control	[?], [?]
11	Zero Trust Network Access (ZTNA)	Provides secure remote access to applications without placing users on the corporate network	Remote Access, VPN Replacement	[?], [?], [?]
12	Behavioral Analytics Integration	Incorporates user and entity behavior analytics (UEBA) to detect anomalies and potential threats	Threat Detection, Monitoring	[?], [?]

investment in monitoring tools and security information and event management (SIEM) systems. Many organizations lack the necessary visibility to effectively implement zero-trust principles.

- 6) **Skills Gap and Expertise Requirements:** Zero-trust architecture requires specialized knowledge spanning network security, identity management, cloud architecture, and threat detection. Organizations face challenges recruiting and retaining personnel with the necessary

expertise.

- 7) **Cost and Resource Constraints:** The financial investment required for zero-trust implementation, including new technologies, tools, training, and potentially redesigning network architecture, presents significant challenges, particularly for smaller enterprises.
- 8) **Third-Party and Supply Chain Complexity:** Extending zero-trust principles to third-party vendors, contractors, and supply chain partners introduces additional

complexity in identity management, access control, and monitoring.

- 9) **Multi-Cloud and Hybrid Environment Challenges:** Organizations operating across multiple cloud providers and hybrid environments face difficulties implementing consistent zero-trust policies and maintaining unified visibility across disparate platforms.
- 10) **Scalability Concerns:** As organizations grow and their IT environments become more complex, maintaining effective zero-trust implementation at scale becomes increasingly challenging, particularly in dynamic environments with frequently changing resources and users.

C. RQ3: Future Directions And Research Gaps

What are the future directions and research gaps in zero-trust architecture?

Zero-trust architecture is an evolving field with significant opportunities for future research and development. Several promising directions and research gaps have been identified:

- 1) **Artificial Intelligence and Machine Learning Integration:** Future research should explore the integration of AI and ML capabilities for automated policy generation, anomaly detection, and adaptive access control. Intelligent systems that can learn normal behavior patterns and automatically adjust security policies based on risk assessment represent a significant opportunity for advancement.
- 2) **Automated Policy Generation and Management:** Developing automated systems for generating, updating, and optimizing access policies based on actual usage patterns and risk profiles would address one of the primary challenges in zero-trust implementation. Research in this area could significantly reduce the operational burden of policy management.
- 3) **Quantum-Safe Zero-Trust Architecture:** As quantum computing advances, current encryption methods face potential vulnerabilities. Research is needed to develop quantum-resistant cryptographic approaches within zero-trust frameworks to ensure long-term security.
- 4) **Zero-Trust for Edge Computing and IoT:** The proliferation of edge computing devices and IoT sensors creates new security challenges. Future research should address how zero-trust principles can be effectively applied to resource-constrained edge devices and IoT ecosystems.
- 5) **Blockchain Integration for Distributed Trust:** Exploring blockchain and distributed ledger technologies for managing trust relationships, audit trails, and access control decisions in zero-trust architectures represents an emerging research area with significant potential.
- 6) **User Experience Optimization:** Research is needed to develop zero-trust implementations that maintain strong security while minimizing friction for legitimate users. Balancing security with usability remains a critical challenge requiring innovative solutions.

- 7) **Zero-Trust Maturity Models and Assessment Frameworks:** Developing standardized maturity models and assessment frameworks would help organizations evaluate their zero-trust implementation progress and identify areas for improvement. Research should focus on creating measurable, practical assessment criteria.
- 8) **Integration with DevSecOps and CI/CD Pipelines:** Future research should address how zero-trust principles can be seamlessly integrated into modern software development practices, including continuous integration and deployment pipelines, to ensure security is built into applications from the start.
- 9) **Privacy-Preserving Zero-Trust Mechanisms:** Developing zero-trust approaches that maintain strong security while preserving user privacy, particularly in the context of increasing privacy regulations, represents an important research direction.
- 10) **Cross-Domain Zero-Trust Architectures:** Research is needed on implementing zero-trust principles across organizational boundaries, enabling secure collaboration while maintaining security controls. This includes multi-party computation, secure enclaves, and federated identity management approaches.

D. RQ4: Key Components And Technologies

What are the key components and technologies used in zero-trust architecture implementations?

Analysis of the 42 selected studies reveals that successful zero-trust implementations typically integrate multiple technology components across different security domains. Identity and Access Management solutions are present in 86% of implementations, making them the most critical component. Network micro-segmentation technologies appear in 71% of cases, while MFA is implemented in 95% of zero-trust architectures. ZTNA solutions have gained significant traction, appearing in 62% of recent implementations, particularly accelerated by remote work requirements. Endpoint security and EDR technologies are present in 79% of implementations, highlighting the importance of device trust verification.

The integration of these components requires careful architectural planning and consideration of interoperability. Organizations typically adopt a phased approach, starting with identity management and MFA, then progressively implementing network segmentation, endpoint security, and advanced monitoring capabilities.

IV. CONCLUSION

To conclude, research on zero-trust architecture continues to evolve as organizations recognize the limitations of traditional perimeter-based security models. We have conducted an SLR of 42 studies from 2018–2024, covering the critical period during which zero-trust principles gained widespread adoption. We have found that micro-segmentation and identity-centric architecture are the most widely implemented design strategies for zero-trust architecture in enterprise software.

TABLE VII
KEY COMPONENTS AND TECHNOLOGIES IN ZERO-TRUST ARCHITECTURE

Component Category	Technologies/Solutions	Description	Ref
Identity and Access Management (IAM)	Okta, Azure AD, Ping Identity, ForgeRock	Centralized identity verification, single sign-on, and access management	[?], [?]
Multi-Factor Authentication (MFA)	Duo Security, RSA SecurID, YubiKey, Google Authenticator	Additional authentication factors beyond passwords	[?], [?]
Network Micro-segmentation	VMware NSX, Cisco ACI, Illumio, Guardicore	Granular network segmentation and policy enforcement	[?], [?]
Software-Defined Perimeter	Appgate SDP, Perimeter 81, Cyxtera	Dynamic perimeter creation based on identity and context	[?], [?]
Zero Trust Network Access (ZTNA)	Zscaler Private Access, Cloudflare Access, Palo Alto Prisma Access	Secure application access without VPN	[?], [?]
Endpoint Detection and Response (EDR)	CrowdStrike Falcon, Carbon Black, SentinelOne	Endpoint security and threat detection	[?], [?]
Cloud Access Security Broker (CASB)	Netskope, McAfee MVISION, Microsoft Cloud App Security	Cloud service security and visibility	[?], [?]
Security Information and Event Management (SIEM)	Splunk, IBM QRadar, LogRhythm	Centralized logging, monitoring, and analytics	[?], [?]
Data Loss Prevention (DLP)	Symantec DLP, Digital Guardian, McAfee Total Protection	Data protection and leakage prevention	[?]
Privileged Access Management (PAM)	CyberArk, BeyondTrust, Thycotic	Management of privileged accounts and access	[?]
API Gateways	Kong, Apigee, AWS API Gateway, Azure API Management	API security and management	[?]
Service Mesh	Istio, Linkerd, Consul Connect	Service-to-service communication security	[?]

Furthermore, we have addressed numerous challenges faced by enterprise architects and security professionals while implementing zero-trust principles. These difficulties primarily include legacy system integration, performance concerns, policy management complexity, and organizational resistance to change. The implementation of zero-trust architecture requires significant investment in technology, training, and process redesign.

Our analysis reveals that successful zero-trust implementations typically integrate multiple technology components, with IAM, MFA, and network segmentation forming the foundational elements. The field shows promise for future research, particularly in areas of AI-driven policy automation, quantum-safe architectures, edge computing security, and user experience optimization.

Moreover, we have identified that automated policy management, privacy-preserving mechanisms, and cross-domain zero-trust architectures represent significant research gaps requiring further investigation. As cyber threats continue to evolve and enterprise environments become increasingly complex, zero-trust architecture will remain a critical area of focus for both academic research and industry implementation.

The transition from traditional security models to zero-trust architecture represents a fundamental shift in how organizations approach cybersecurity. While challenges remain, the benefits of reduced attack surface, improved visibility, and enhanced security posture make zero-trust architecture an essential consideration for modern enterprise software systems.

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